

Six Trigonometric Ratios & the Pythagorean Theorem

Trigonometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Use the Pythagorean Theorem to find a missing side of a right triangle before applying trig ratios
- Evaluate all six trigonometric ratios (sin, cos, tan, csc, sec, cot) for a given angle in a right triangle
- Convert between degrees and radians and evaluate the six trig functions for special angles using the unit circle

Problems

1. A right triangle has legs of length 3 and 4. Find the hypotenuse using the Pythagorean Theorem.

$$c^2 = 3^2 + 4^2$$

2. A right triangle has a hypotenuse of 10 and one leg of 6. Find the missing leg using the Pythagorean Theorem.

$$10^2 = 6^2 + x^2$$

3. Using the triangle with hypotenuse 2, one leg 1, and the other leg equal to the square root of 3, find $\sin 30^\circ$.

$$\sin 30^\circ = \frac{\text{opposite}}{\text{hypotenuse}}$$

4. Using the same 30-60-90 triangle (sides 1, square root of 3, and 2), find $\cos 30^\circ$ and $\tan 30^\circ$. Rationalize $\tan 30^\circ$.

$$\cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}}$$

5. Find the three reciprocal trig ratios (csc, sec, and cot) for 30° using the 30-60-90 triangle.



$$\csc 30^\circ = \frac{1}{\sin 30^\circ}, \quad \sec 30^\circ = \frac{1}{\cos 30^\circ}, \quad \cot 30^\circ = \frac{1}{\tan 30^\circ}$$

6. Convert the angle π over 3 radians into degrees, then state the coordinates of that angle on the unit circle.

$$\frac{\pi}{3} \times \frac{180^\circ}{\pi}$$

7. Using the unit circle coordinates for π over 3, evaluate all six trig functions for θ equals π over 3.

$$\theta = \frac{\pi}{3}, \quad (x, y) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

8. A right triangle has a hypotenuse of 5 and one leg of 2. First find the missing leg, then evaluate all six trig ratios for the acute angle opposite the leg of length 2. Rationalize all answers.

$$c = 5, \quad a = 2, \quad b = ?$$

9. Given that $\sin(\theta)$ equals 5 over 13 and θ is in the first quadrant, use the Pythagorean identity to find $\cos(\theta)$, then evaluate all six trig functions.

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \sin \theta = \frac{5}{13}$$

10. A right triangle is placed in the coordinate plane with angle θ at the origin. The terminal side passes through the point (negative 3, 4). Find all six trig functions of θ , rationalizing where necessary.

$$r = \sqrt{(-3)^2 + 4^2}$$



Six Trigonometric Ratios & the Pythagorean Theorem — Answer Key

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Answer Key

1. Answer: $c = 5$

- $c^2 = 9 + 16 = 25$
- $c = \sqrt{25} = 5$

2. Answer: $x = 8$

- $100 = 36 + x^2$
- $x^2 = 64$, so $x = 8$

3. Answer: $\sin 30^\circ = 1/2$

- The side opposite 30° is 1, and the hypotenuse is 2
- $\sin 30^\circ = 1/2$

4. Answer: $\cos 30^\circ = \sqrt{3}/2$; $\tan 30^\circ = \sqrt{3}/3$

- $\cos 30^\circ = \text{adjacent/hypotenuse} = \sqrt{3}/2$
- $\tan 30^\circ = \text{opposite/adjacent} = 1/\sqrt{3} = \sqrt{3}/3$ after rationalizing

5. Answer: $\csc 30^\circ = 2$; $\sec 30^\circ = 2\sqrt{3}/3$; $\cot 30^\circ = \sqrt{3}$

- $\csc 30^\circ = 1/(1/2) = 2$
- $\sec 30^\circ = 1/(\sqrt{3}/2) = 2/\sqrt{3} = 2\sqrt{3}/3$ after rationalizing; $\cot 30^\circ = 1/(\sqrt{3}/3) = \sqrt{3}$

6. Answer: 60° ; coordinates $(1/2, \sqrt{3}/2)$

- $(\pi/3) \times (180^\circ/\pi) = 60^\circ$
- On the unit circle, 60° has coordinates $(\cos 60^\circ, \sin 60^\circ) = (1/2, \sqrt{3}/2)$

7. Answer: $\sin=\sqrt{3}/2$, $\cos=1/2$, $\tan=\sqrt{3}$, $\csc=2\sqrt{3}/3$, $\sec=2$, $\cot=\sqrt{3}/3$

- $\sin(\pi/3)=y=\sqrt{3}/2$; $\cos(\pi/3)=x=1/2$; $\tan(\pi/3)=y/x=\sqrt{3}$
- $\csc(\pi/3)=1/y=2/\sqrt{3}=2\sqrt{3}/3$; $\sec(\pi/3)=1/x=2$; $\cot(\pi/3)=x/y=1/\sqrt{3}=\sqrt{3}/3$

8. Answer: $b=\sqrt{21}$; $\sin=2/5$, $\cos=\sqrt{21}/5$, $\tan=2/\sqrt{21}=2\sqrt{21}/21$, $\csc=5/2$, $\sec=5/\sqrt{21}=5\sqrt{21}/21$, $\cot=\sqrt{21}/2$

- $b^2 = 5^2 - 2^2 = 25 - 4 = 21$, so $b = \sqrt{21}$
- $\sin=2/5$; $\cos=\sqrt{21}/5$; $\tan=2/\sqrt{21}=2\sqrt{21}/21$; $\csc=5/2$; $\sec=5/\sqrt{21}=5\sqrt{21}/21$; $\cot=\sqrt{21}/2$

9. Answer: $\cos=12/13$, $\tan=5/12$, $\csc=13/5$, $\sec=13/12$, $\cot=12/5$

- $\cos^2\theta = 1 - (5/13)^2 = 1 - 25/169 = 144/169$, so $\cos \theta = 12/13$
- $\tan=5/12$; $\csc=13/5$; $\sec=13/12$; $\cot=12/5$

10. Answer: $r=5$; $\sin=4/5$, $\cos=-3/5$, $\tan=-4/3$, $\csc=5/4$, $\sec=-5/3$, $\cot=-3/4$





- $\sin=y/r=4/5$; $\cos=x/r=-3/5$; $\tan=y/x=-4/3$; $\csc=5/4$; $\sec=-5/3$; $\cot=-3/4$

